Only Coding

📘 DSA & Java Practice Roadmap (Organized from Personal Notes)

📖 Objective

The goal is to organize all coding problems topic-wise to enable structured preparation for Data Structures & Algorithms (DSA).
Initial focus was on Java Collection Framework (JCF) to build a strong base for solving these problems.

🔵 1. Pattern / Loop-Based Problems (Beginner Level)

Purpose: Build logic using loops

- Pyramid Pattern
- Diamond Pattern
- Left Triangle Pattern
- Right Triangle Pattern

🔵 2. String Problems

🔹 **Basic**

- Reverse string (with / without built-in methods)
- Palindrome check
- Prove string immutability

🔹 **Frequency-Based (Using HashMap)**

- Duplicate characters in string (e.g., "Mississippi")
- Distinct character count
- Most frequent character

🔹 **Advanced**

- Anagram check
- Longest substring

- Count palindromic substrings

🟦 3. Array / List Problems (Core DSA)

◆ **Basic**

- Find max / min element
- Second largest element
- Reverse array
- Rotate array by k

◆ **Intermediate**

- Merge two arrays (without duplicates)
- Split array into chunks
- Find missing number
- Find common elements

◆ **Advanced**

- Triplet sum problem
- Longest consecutive sequence
- Sliding window maximum

🟦 4. HashMap / HashSet Problems

◆ **Frequency Problems**

- Count occurrences of elements
- Distinct element count
- Most frequent element

◆ **Set-Based Problems**

- Remove duplicates
- Union of arrays
- Intersection of arrays
- First repeated element

- **Advanced**
  - Top K frequent elements
  - Graph adjacency list using HashMap

5. Java 8 / Stream API

- **Basic**
  - Convert collection to stream
  - Sorting
  - Filtering
- **Intermediate**
  - Sum and average
  - Find longest string
  - Distinct elements
- **Advanced**
  - Group employees by department
  - Nested grouping
  - FlatMap problems

6. Stack / Queue

- Parenthesis balancing
- Stack implementation using arrays
- BFS / DFS basics

7. Linked List

- Reverse linked list
- Detect loop in linked list

◍ 8. Trees

- Tree traversal (Inorder, Preorder, Postorder)

○ 9. Graph

- Adjacency list (using HashMap)
- BFS and DFS traversal

◍ 10. Heap / Advanced Problems

- Sliding window maximum
- Kth largest element

🧠 Key Insight

The difficulty in solving problems is not due to weak concepts, but due to lack of structured progression.

🚀 Study Plan (Recommended Approach)

Phase 1: Foundation (Current Stage)

- ArrayList
- HashSet
- HashMap

Focus on:

- Frequency problems
- Duplicate handling
- Union / Intersection
- Missing number
- Longest consecutive sequence

Phase 2: Core DSA (Array + String)

- Two Sum
- Anagram
- Sliding window basics
- Subarray problems

Phase 3: Intermediate Data Structures

- Stack
- Queue

Phase 4: Advanced Structures

- Linked List
- Trees